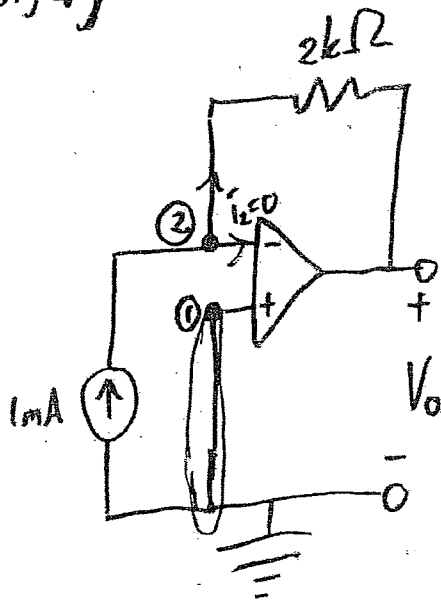


5.8.) a.)



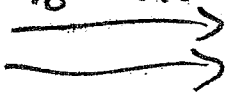
1) $V_1 = V_2 = 0V$

2) KCL at node (2):

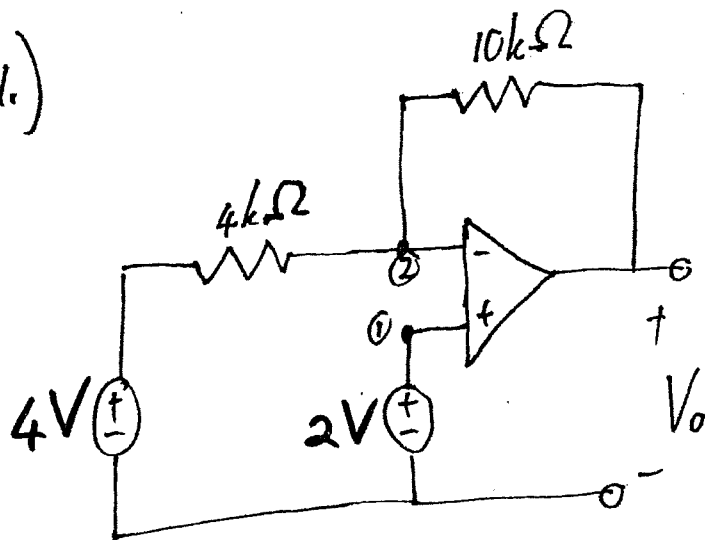
$$0,001 = \frac{V_2 - V_o}{2000}$$

$$2 = (0) - V_o$$

$$\underline{V_o = -2V}$$



5.21.)



① $V_1 = V_2 = 2V$

② KCL at node 2:

$$\frac{V_2 - 4}{4000} + \frac{V_2 - V_o}{10000} = 0$$

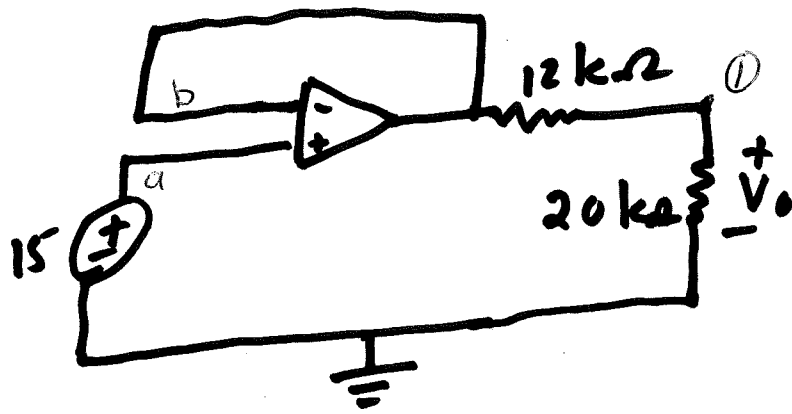
$$\frac{2 - 4}{4000} + \frac{2 - V_o}{10000} = 0$$

$$-10 + 2 - 2V_o = 0$$

$$-2V_o = 8$$

$$\underline{\underline{V_o = -3V}}$$

5.25)



Find V_o

$$V_a = V_b = 15V$$

KCL and Node 1:

$$\frac{V_b - V_o}{12k} = \frac{V_o}{20k}$$

$$\frac{15 - V_o}{12k} = \frac{V_o}{20k}$$

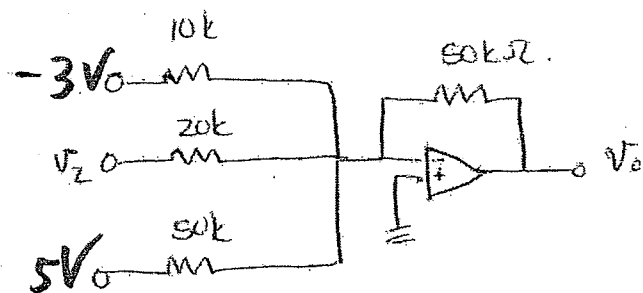
$$(15 - V_o)(5) = 3V_o$$

$$75 - 5V_o = 3V_o$$

$$75 = 8V_o$$

$$\therefore V_o = 9.375V$$

5.39

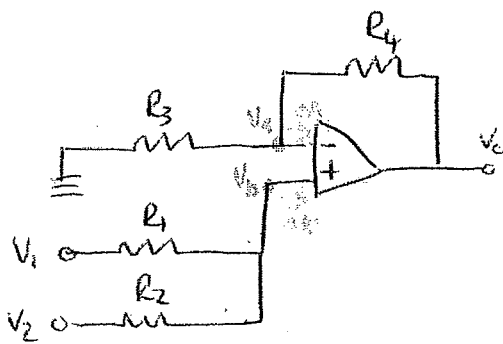
Determine V_2 if $V_0 = -16,5V$

$$-16,5 = -\left[\left(\frac{50}{10}\right)(-3) + \left(\frac{50}{20}\right)V_2 + \left(\frac{50}{50}\right)(5)\right]$$

$$-16,5 - 15 + 5 = -2,5V_2$$

$$V_2 = 10,6V \rightarrow$$

5.44



Show that output voltage is:

$$V_o = \frac{(R_3 + R_4)}{R_3(R_1 + R_2)} (R_2 V_1 + R_1 V_2)$$

* $V_a = V_b$ ——— ①

* KCL @ V_a :

$$\frac{0 - V_a}{R_3} = \frac{V_a - V_o}{R_4}$$

$$-V_a = \frac{V_a R_3}{R_4} - \frac{V_o R_3}{R_4}$$

$$-V_a \left(1 + \frac{R_3}{R_4}\right) = -\left(\frac{R_3}{R_4}\right) V_o$$

$$V_a = \left[\frac{\frac{R_3}{R_4}}{1 + \left(\frac{R_3}{R_4}\right)} \right] V_o$$

$$= \left[\frac{1}{\left(\frac{R_4}{R_3} + 1\right)} \right] V_o \quad \text{————— ②}$$

* KCL @ V_b

$$\frac{V_1 - V_b}{R_1} + \frac{V_2 - V_b}{R_2} = 0$$

$$R_2 V_1 - R_2 V_b + (R_1 V_2 - R_1 V_b) = 0$$

$$V_b (R_1 + R_2) = (R_2 V_1 + R_1 V_2)$$

$$V_b = \frac{(R_2 V_1 + R_1 V_2)}{R_1 + R_2} \quad \text{————— ③}$$

→ EOP

set $V_a = V_b$.

$$\Rightarrow \frac{V_o}{\frac{R_4}{R_3} + 1} = \frac{(R_2 V_1 + R_1 V_2)}{R_1 + R_2}$$

$$V_o = (R_2 V_1 + R_1 V_2) \left[\frac{\left(\frac{R_4}{R_3} + 1 \right) R_3}{(R_1 + R_2)} \right]$$

$$= (R_2 V_1 + R_1 V_2) \left[\frac{R_4 + R_3}{R_3 (R_1 + R_2)} \right]$$

Answers

$$5.8a) -2V ; b) V_0 = -1V$$

$$5.21) V_0 = -3V$$

$$5.23) \frac{V_0}{V_s} = -\frac{R_f}{R_1}$$

$$5.33) i_{x_2} = -5\text{ mA} ; P_{3k\Omega} = 75\text{ mW}$$

$$5.37) V_0 = -20.6V$$

$$5.39) V_2 = 10.6V$$

$$5.44) \text{Proof}$$

$$5.47) V_0 = 14.09V$$